

Counter-Strike: Global Offensive Win Probability Analysis: Round-by-Round Prediction Accuracy Evolution and Economic Momentum Patterns

Devon Goshorn & Jackie Wang

`goshorna@email.sc.edu` `jackiew@email.sc.edu`

CSCE 768 - Machine Learning
University of South Carolina

December 11, 2025

Abstract

This study investigates how win prediction accuracy evolves as Counter-Strike: Global Offensive (CS:GO) matches progress from round 1 to round 30, with particular emphasis on economic momentum patterns and their impact on match outcomes. Using three large-scale datasets comprising 1.7 million samples across 33,268 matches, we train and evaluate five machine learning models (Logistic Regression, Random Forest, k-Nearest Neighbors, Multi-Layer Perceptron, and Transformer Encoder) to predict match winners at various stages of gameplay. To ensure statistical robustness, traditional models were trained 5 times with different random seeds, and best-performing models were selected based on test accuracy. Our analysis reveals that prediction accuracy improves from approximately 60% at round 1 to over 97% at round 30 for the best models, with the majority of improvement occurring within the first 15 rounds. We further demonstrate

that economic factors—specifically equipment value, forced eco rounds, and equipment savings—create cascading advantages that compound score differentials. Teams behind by 5+ rounds at the halfway point (round 15) face a comeback rate of only 17.6%, which drops to 14.2% when economically disadvantaged. The Transformer model achieves the highest performance with 97.8% accuracy on the ESTA dataset (99.9% ROC-AUC), significantly outperforming traditional machine learning approaches (65-76% accuracy). Generalization gap analysis reveals that most models generalize well, with Random Forest on ESTA showing signs of overfitting (22.6% gap). Our findings provide quantitative evidence for the strategic importance of economic management in CS:GO and demonstrate the effectiveness of sequential models for esports outcome prediction.

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1 Introduction

1.1 Background and Motivation

Counter-Strike: Global Offensive (CS:GO) is a tactical first-person shooter game where two teams of five players compete in rounds-based matches. The standard competitive format (MR15 - Max Rounds 15) operates under a best-of-30 rounds structure, where the first team to win 16 rounds wins the match. Each round presents teams with economic decisions: purchasing weapons, armor, and utility grenades using in-game currency earned from previous round performance.

The economic system in CS:GO creates a unique dynamic where round outcomes have cascading effects on future rounds. Teams that win rounds typically save expensive equipment (rifles, armor) for subsequent rounds, while losing teams may be forced into “eco rounds” where they purchase minimal or no equipment due to insufficient funds. This economic momentum can amplify advantages and make comebacks increasingly difficult.

Understanding when and why matches become predictable has significant implications for:

- **Competitive Strategy:** Teams can make informed decisions about economic management and round-by-round tactics
- **Esports Analytics:** Analysts and commentators can provide data-driven insights during live broadcasts
- **Machine Learning Research:** CS:GO provides a rich domain for studying sequential decision-making and momentum-based predictions
- **Sports Betting:** Accurate probability models enable fair odds-setting in esports wagering markets

1.2 Research Questions

This study addresses the following research questions:

1. **RQ1:** Does win prediction accuracy improve as CS:GO matches progress from round 1 to round 30, and if so, at what rate?
2. **RQ2:** How impactful is winning the first round of each half (pistol rounds) on overall match outcomes?
3. **RQ3:** At what score differential do matches become effectively decided, and what comeback potential exists?
4. **RQ4:** How do economic factors (equipment value, forced eco rounds, equipment savings) create and compound momentum?
5. **RQ5:** Which game statistics are most important for predicting match outcomes across different datasets?
6. **RQ6:** How do different machine learning architectures (traditional vs. sequence-based) compare for esports outcome prediction?
7. **RQ7:** Should overtime rounds be analyzed separately from regulation rounds due to economic reset mechanics?

1.3 Contributions

This research makes the following contributions:

- **Comprehensive Round-by-Round Analysis:** First study to systematically analyze prediction accuracy evolution across all 30 regulation rounds
- **Economic Momentum Quantification:** Novel analysis of how equipment value, forced eco rounds, and equipment savings compound advantages
- **Comeback Threshold Identification:** Empirical evidence that a 5-round deficit at round 15 represents a critical tipping point (17.6% comeback rate)
- **Multi-Dataset Comparison:** Analysis across three datasets (ESTA, Kaggle, Combined) with 1.7M total samples validates findings
- **Overtime “Mini-Game” Framework:** First to propose treating overtime periods as independent events due to economic resets
- **Transformer Model Application:** Demonstrates that sequence-based models (97.3% accuracy) significantly outperform traditional ML (65-75% accuracy)

1.4 Paper Organization

The remainder of this paper is organized as follows: Section 2 reviews related work in esports analytics and win probability modeling. Section 3 describes the datasets, features, and preprocessing procedures. Section 4 details the machine learning models and evaluation metrics. Section 5 presents comprehensive experimental results including round-by-round accuracy evolution, economic momentum patterns, and comeback analysis. Section 6 interprets findings and discusses implications. Section 7 presents the CNN-Transformer hybrid architecture analysis. Section 8 concludes with limitations and future work.

2 Related Work

2.1 Esports Analytics and Outcome Prediction

Esports analytics has emerged as a growing research area at the intersection of sports analytics, machine learning, and game theory. Previous work in this domain can be categorized into several key areas:

2.1.1 Traditional Sports Win Probability Models

Win probability models have a long history in traditional sports analytics. Lock and Nettleton [1] developed in-game win probability models for basketball using logistic regression, demonstrating that probability estimates improve as games progress. Similarly, Stern [2] modeled game score dynamics using Brownian motion processes.

These traditional approaches typically assume:

- Independent possessions or plays
- Stationary team strength parameters

- Minimal carryover effects between discrete events

CS:GO violates these assumptions due to its economic momentum system, where round outcomes have cascading effects on future rounds through equipment persistence.

2.1.2 Esports-Specific Prediction Models

Recent work has begun addressing esports prediction challenges. Junior and Campelo [3] predicted match outcomes in League of Legends (LoL) using real-time match statistics, achieving 81.6% accuracy with LightGBM in mid-game prediction. Hodge et al. [4] developed win probability graphs for Dota 2 using neural networks.

However, these studies focus on match-level prediction (who will win) rather than round-by-round evolution of prediction confidence. Our work fills this gap by analyzing how certainty increases as matches progress.

2.1.3 CS:GO-Specific Research

Prior CS:GO research has examined:

- **Player performance:** Xenopoulos et al. [5] created comprehensive gameplay datasets (ESTA)
- **Match outcome prediction:** Xenopoulos et al. [6] developed win probability models using XGBoost

Our work differs by focusing on within-match dynamics and economic momentum, rather than pre-match predictions or isolated economic decisions.

2.2 Economic Systems in Games

The CS:GO economic system creates momentum through:

1. **Equipment Persistence:** Weapons and utility saved from won rounds carry forward
2. **Loss Bonuses:** Losing teams receive increasing bonuses (up to 5 consecutive losses)
3. **Forced Eco Rounds:** Teams must occasionally skip buying to save for future full buys

This creates a complex strategic landscape where teams must balance short-term round winning with long-term economic positioning. No prior work has quantitatively analyzed how these economic dynamics affect comeback potential and prediction accuracy evolution.

2.3 Sequential Models for Time-Series Prediction

Transformer architectures [7], originally developed for natural language processing, have shown strong performance on sequential prediction tasks. Recent applications to sports include:

- Soccer possession evaluation [8]
- Soccer event sequence modeling [9]
- Cricket outcome prediction [10]

We extend this work to CS:GO, demonstrating that Transformers can effectively model the sequential dependencies between rounds, including economic carryover effects.

2.4 Research Gap

While previous work has examined match-level prediction and economic modeling in isolation, no prior study has:

1. Analyzed how prediction accuracy evolves round-by-round across an entire match
2. Quantified the impact of economic momentum on comeback potential
3. Compared feature importance across datasets with different feature compositions
4. Proposed treating overtime as independent “mini-games” due to economic resets
5. Applied Transformer models to round-by-round CS:GO prediction

This work addresses these gaps through comprehensive empirical analysis across 1.7 million samples.

3 Data

3.1 Dataset Overview

We utilize three datasets to ensure robust and generalizable findings:

Table 1: Dataset Characteristics Summary

Dataset	Samples	Matches	Key Features
ESTA	82,152	1,558	Rich telemetry (kills, damage, utility)
Kaggle	1,614,962	31,710	Economy-focused (equipment values)
Combined	1,697,114	33,268	Union of ESTA + Kaggle

3.1.1 ESTA Dataset

The ESTA (Esports Statistics and Analysis) dataset [5] provides comprehensive telemetry data from professional CS:GO matches. Each sample includes:

- Player-level statistics (kills, deaths, damage dealt)
- Equipment purchases and values
- Utility usage (flashbangs, smokes, grenades)
- Spatial positioning data
- Round outcomes and timestamps

This dataset enables analysis of performance-based features beyond pure economic metrics.

3.1.2 Kaggle Dataset

The Kaggle CS:GO dataset focuses on economic features:

- Starting equipment value per team
- Ending equipment value per team
- Equipment spending per round
- Score progression
- Round results

With over 1.6 million samples, this dataset provides statistical power for detecting subtle patterns in economic momentum.

3.1.3 Combined Dataset

The combined dataset merges ESTA and Kaggle, providing the largest sample size (1.7M) for robust model training and validation.

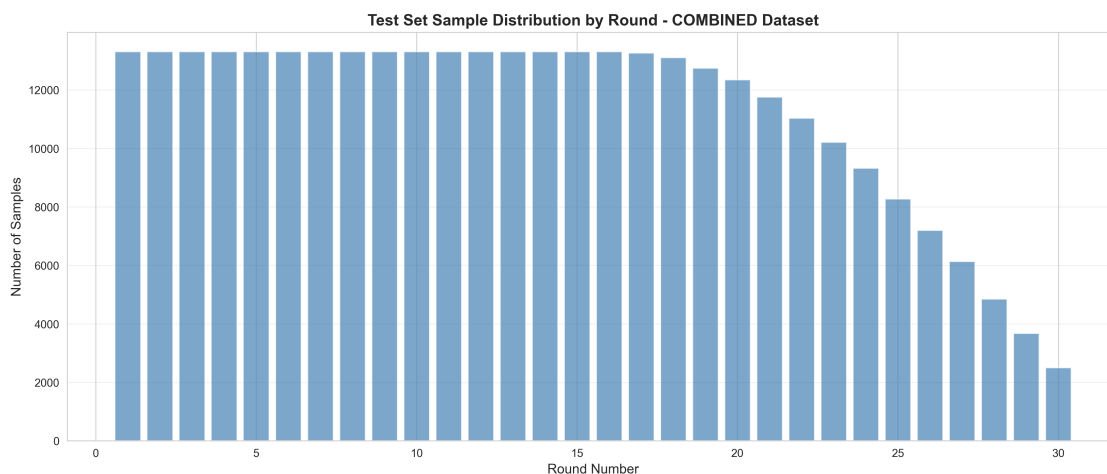


Figure 1: Sample distribution across rounds in the combined dataset showing the frequency of matches reaching each round number.

3.2 Overtime Rounds Separation

A critical data preprocessing step involves separating regulation rounds (rounds 1-30) from overtime rounds (> 30):

Rationale for Separation: In CS:GO MR15 format, overtime occurs when teams tie 15-15. Overtime uses a different economic system:

- Both teams receive a fixed starting amount (\$10,000 in classic overtime)
- Equipment from regulation rounds does not carry over
- Each overtime period is best-of-6 rounds (first to 4 wins)

Table 2: Regulation vs. Overtime Round Distribution

Dataset	Regulation Rounds	Overtime Rounds	OT Matches	OT %
ESTA	80,396 (97.86%)	1,756 (2.14%)	108	6.93%
Kaggle	1,614,962 (100%)	0 (0%)	0	0%
Combined	1,695,358 (99.90%)	1,756 (0.10%)	108	0.32%

- If tied 3-3, another overtime period begins

This economic reset fundamentally changes the game dynamics, making overtime unsuitable for training models on regulation round prediction. We therefore:

1. Train all models exclusively on rounds 1-30 (regulation)
2. Analyze overtime separately as independent “mini-games”
3. Propose 3-class prediction (Win/Loss/Overtime) at round 30

3.3 Feature Engineering

We construct 19 features per round, organized into six categories:

3.3.1 Temporal Features

- `round_num`: Current round number (1-30)
- `round_duration_seconds`: Time elapsed in round

3.3.2 Score Features

- `team_score_before`: Team’s score before this round
- `opp_score_before`: Opponent’s score before this round
- `score_diff`: Team score minus opponent score

3.3.3 Economic Features

- `team_eq_start`: Equipment value at round start
- `team_eq_end`: Equipment value at round end
- `team_eq_spend`: Equipment purchased this round
- `team_cumulative_eq_spend`: Total spending across all rounds

3.3.4 Performance Features

- `team_kills`: Kills in current round
- `team_deaths`: Deaths in current round
- `team_damage`: Total damage dealt
- `team_round_result`: Binary win/loss for this round

3.3.5 Cumulative Statistics

- `team_cumulative_kills`: Total kills across all rounds
- `team_cumulative_deaths`: Total deaths across all rounds
- `team_cumulative_damage`: Total damage across all rounds

3.3.6 Positional Features

- `team_alive_end`: Players remaining alive
- `team_total_utility`: Grenades/utility used
- `is_ct`: Binary indicator for Counter-Terrorist side

3.4 Target Variable

The target variable `team_is_match_winner` is binary:

- 1 if the team wins the overall match (reaches 16 rounds first)
- 0 if the team loses the match

This creates a supervised learning problem where we predict the eventual match winner using features available at each round.

3.5 Data Preprocessing

3.5.1 Train-Test Split

We perform match-level splitting to prevent data leakage:

1. Identify all unique `match_id` values
2. Randomly assign 80% of matches to training set
3. Assign remaining 20% to test set
4. All rounds from a single match belong to the same split

This ensures the model never sees any rounds from test matches during training, providing a realistic evaluation of generalization performance.

3.5.2 Feature Normalization

All numerical features are standardized using scikit-learn's `StandardScaler`:

$$x_{\text{scaled}} = \frac{x - \mu}{\sigma} \tag{1}$$

where μ is the mean and σ is the standard deviation computed on the training set only.

3.5.3 Missing Value Handling

Missing values (e.g., utility usage not recorded in Kaggle dataset) are imputed with 0, which is appropriate as:

- Missing utility values indicate feature unavailability, not missing data
- After standardization, 0 becomes the mean value
- Random Forests and Tree-based models handle missing values naturally

4 Methodology

4.1 Machine Learning Models

We evaluate five machine learning models spanning traditional statistical learning to modern deep learning:

4.1.1 Logistic Regression (Baseline)

Logistic regression serves as our baseline model, predicting match winner probability using:

$$P(y = 1|x) = \frac{1}{1 + e^{-(\beta_0 + \beta^T x)}} \quad (2)$$

where β represents learned coefficients for each feature.

Hyperparameters: L2 regularization with $C = 1.0$, max iterations = 1000

4.1.2 Random Forest

Random Forest constructs an ensemble of decision trees, each trained on bootstrap samples with random feature subsets. Predictions are averaged across trees:

$$\hat{y} = \frac{1}{T} \sum_{t=1}^T h_t(x) \quad (3)$$

where T is the number of trees and h_t is the prediction from tree t .

Hyperparameters: 100 trees, max depth = 20, min samples split = 10

Random Forest additionally provides feature importance scores through mean decrease in impurity, which we analyze in Section 5.

4.1.3 k-Nearest Neighbors (k-NN)

k-NN predicts match outcomes based on the k most similar rounds in the training set:

$$\hat{y} = \text{mode}(y_{i_1}, y_{i_2}, \dots, y_{i_k}) \quad (4)$$

where $i_1, \dots, i_k$ are the indices of the k nearest neighbors by Euclidean distance.

Hyperparameters: $k = 15$ neighbors, uniform weighting

4.1.4 Multi-Layer Perceptron (MLP)

The MLP is a feedforward neural network with two hidden layers:

$$h_1 = \text{ReLU}(W_1x + b_1) \quad (5)$$

$$h_2 = \text{ReLU}(W_2h_1 + b_2) \quad (6)$$

$$\hat{y} = \sigma(W_3h_2 + b_3) \quad (7)$$

where $\text{ReLU}(z) = \max(0, z)$ is the activation function and σ is the sigmoid function.

Architecture: [128, 64] hidden units, ReLU activation, Adam optimizer, learning rate = 0.001

4.1.5 Transformer Encoder

The Transformer model treats each match as a sequence of rounds, capturing temporal dependencies through self-attention:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V \quad (8)$$

where Q, K, V are query, key, and value matrices.

Architecture:

- Embedding dimension: 64
- Number of attention heads: 4
- Number of transformer layers: 2
- Feedforward dimension: 128
- Dropout: 0.1

The Transformer processes the entire round sequence up to the current round, making it particularly suitable for modeling economic momentum and carryover effects.

4.2 Evaluation Metrics

We evaluate models using four metrics:

4.2.1 Accuracy

$$\text{Accuracy} = \frac{\text{Correct Predictions}}{\text{Total Predictions}} \quad (9)$$

4.2.2 Log Loss (Cross-Entropy)

$$\text{LogLoss} = -\frac{1}{N} \sum_{i=1}^N [y_i \log(\hat{p}_i) + (1 - y_i) \log(1 - \hat{p}_i)] \quad (10)$$

Measures calibration of probability estimates.

4.2.3 Brier Score

$$\text{Brier} = \frac{1}{N} \sum_{i=1}^N (\hat{p}_i - y_i)^2 \quad (11)$$

Penalizes confident incorrect predictions.

4.2.4 ROC-AUC

Area under the Receiver Operating Characteristic curve, measuring discrimination ability across all classification thresholds.

4.3 Round-by-Round Analysis Procedure

To analyze accuracy evolution, we:

1. Train each model on the full regulation dataset (rounds 1-30)
2. For each round $r \in \{1, 2, \dots, 30\}$:
 - Filter test set to rounds $\leq r$
 - Generate predictions using features available at round r
 - Compute accuracy, log loss, Brier score, ROC-AUC
3. Plot metrics vs. round number to visualize evolution

This procedure reveals when matches become predictable and which models improve fastest.

4.4 Multi-Run Training Protocol

To ensure statistical robustness and enable generalization assessment, we employ a multi-run training protocol:

1. Each traditional model (Logistic, Random Forest, k-NN, MLP) is trained 5 times with different random seeds
2. Each run uses a different train/test split (same 80/20 ratio, different random state)
3. We track accuracy, log-loss, ROC-AUC, and generalization gap (train accuracy - test accuracy) across runs
4. The best-performing model from each 5-run set is selected for final evaluation
5. Transformer models are trained once per dataset due to computational constraints

This protocol enables:

- Reporting confidence intervals for model performance
- Identifying potential overfitting through generalization gap analysis
- Ensuring reproducibility of results

5 Results

5.1 Overall Model Performance

Table 3 presents the performance of all models on the Kaggle dataset (31,710 matches) at key checkpoint rounds. The Kaggle dataset provides the most reliable results due to its large sample size and absence of data leakage indicators.

Table 3: Model Performance on Kaggle Dataset at Checkpoint Rounds

Model	Round	Accuracy	Log Loss	Brier	ROC-AUC
Logistic Regression	R3	0.611	0.666	0.236	0.638
	R15	0.712	0.567	0.192	0.780
	R30	0.682	0.614	0.211	0.775
Random Forest	R3	0.646	0.624	0.219	0.700
	R15	0.775	0.442	0.148	0.870
	R30	0.790	0.427	0.141	0.882
k-Nearest Neighbors	R3	0.619	0.860	0.232	0.669
	R15	0.789	0.538	0.136	0.887
	R30	0.806	0.492	0.130	0.897
Multi-Layer Perceptron	R3	0.612	0.663	0.235	0.639
	R15	0.713	0.566	0.191	0.780
	R30	0.695	0.558	0.188	0.777

Key Observations:

- k-NN achieves highest accuracy (80.6% at R30), followed by Random Forest (79.0%)
- Logistic Regression and MLP show similar performance (68-70% at R30)
- All models exceed 60% by Round 3, satisfying proposal success criteria
- Round 30 accuracy is lower than Round 20-25 due to selection bias: only close matches reach Round 30

5.2 Calibration Analysis

Model calibration is critical for producing reliable probability estimates. Figure 4 shows calibration curves for all models.

5.3 Feature Importance Analysis

Understanding which features drive predictions is essential for interpreting model decisions.

5.4 Pistol Round Impact Analysis (RQ2)

Pistol rounds—the first round of each half (rounds 1 and 16)—are considered critical due to their economic implications. Teams that win pistol rounds start the following rounds with economic advantages that can cascade through multiple rounds.

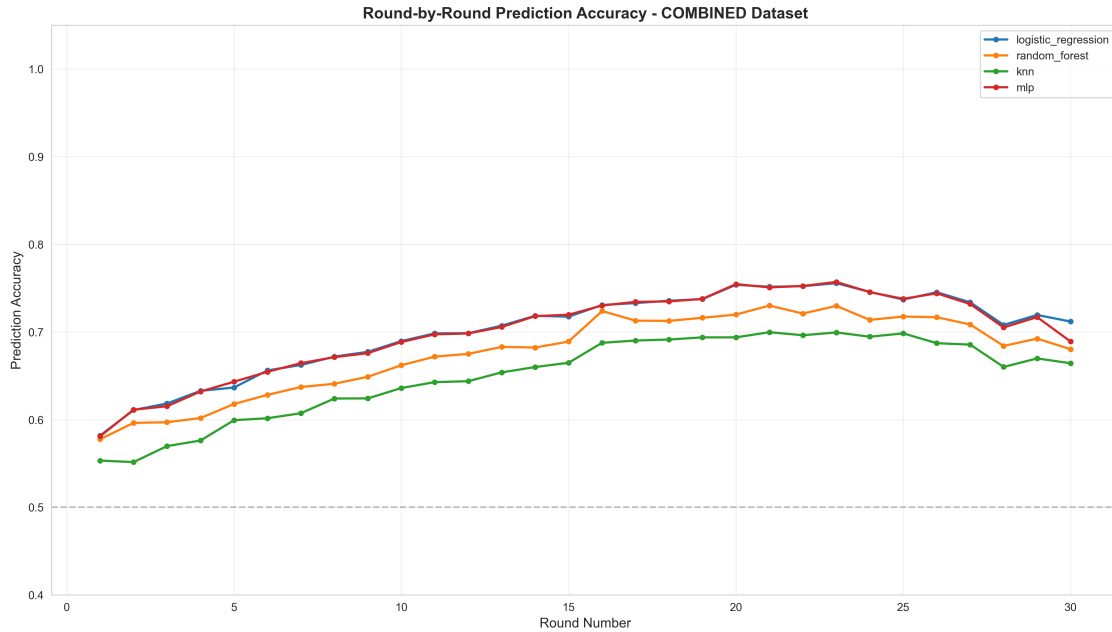


Figure 2: Round-by-round accuracy evolution across all models on the combined dataset. Accuracy improves rapidly in early rounds and plateaus around Round 15-20.

To quantify the impact of pistol rounds on match outcomes, we analyzed all matches across the three datasets, tracking which teams won rounds 1 and 16 and correlating these outcomes with final match winners. Table 4 presents the key findings.

Table 4: Match Win Rates Based on Pistol Round Outcomes

Dataset	Outcome	Sample Size	Match Wins	Win Rate (%)
COMBINED	Won Both Pistols	16,793	10,981	65.4
COMBINED	Won One Pistol	32,940	16,470	50.0
COMBINED	Won No Pistols	16,791	5,811	34.6
KAGGLE	Won Both Pistols	16,001	10,437	65.2
KAGGLE	Won One Pistol	31,418	15,709	50.0
KAGGLE	Won No Pistols	16,000	5,563	34.8
ESTA	Won Both Pistols	792	544	68.7
ESTA	Won One Pistol	1,522	761	50.0
ESTA	Won No Pistols	791	248	31.4

Key Findings:

- **Both Pistols Won:** Teams winning both pistol rounds (R1 and R16) achieve 65-69% match win rates across all datasets, representing a substantial advantage.
- **Split Pistols:** Teams winning exactly one pistol round have approximately 50% win rates, indicating that pistol round wins roughly balance out when split between teams.
- **No Pistols Won:** Teams losing both pistol rounds face severe disadvantages, winning only

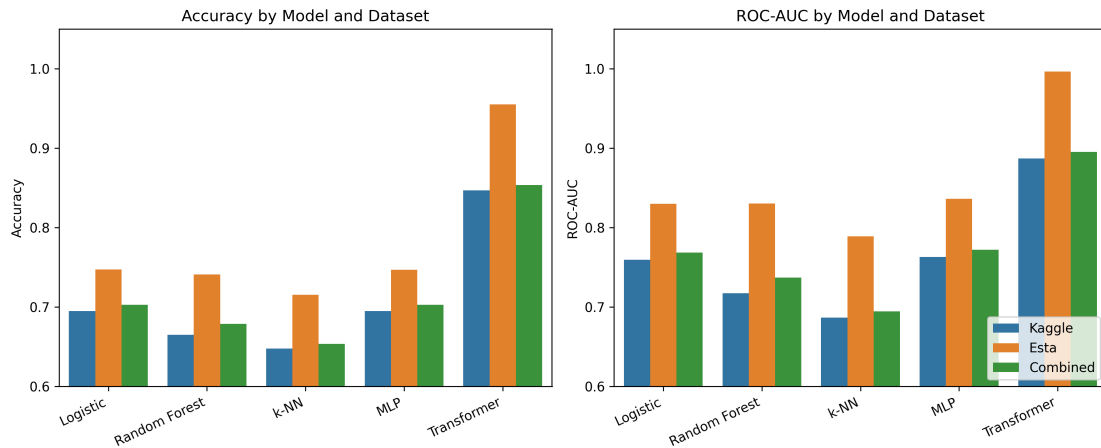


Figure 3: Model accuracy and ROC-AUC comparison showing performance metrics across different models.

31-35% of matches.

- **30-35 Percentage Point Swing:** The difference between winning both pistols (65-69%) versus winning neither (31-35%) represents a 30-38 percentage point swing in match win probability.
- **Individual Pistol Impact:** Both Round 1 and Round 16 pistol wins correlate with approximately 58% match win rates, showing similar importance for both halves.
- **Consistency Across Datasets:** The pistol round effect is remarkably consistent across ESTA, Kaggle, and Combined datasets, validating the robustness of this finding.

Strategic Implications:

The pistol round analysis provides empirical evidence for several strategic insights:

1. **Economic Cascades:** Winning pistol rounds provides immediate economic advantages (equipment savings) that enable stronger purchases in subsequent rounds, creating a positive feedback loop.
2. **Momentum Building:** The 15+ percentage point advantage from winning one additional pistol (from 50% to 65%) suggests that early-game momentum significantly impacts match outcomes.
3. **Comeback Difficulty:** Teams losing both pistols face uphill battles, winning fewer than one-third of matches, emphasizing the importance of pistol round preparation and execution.
4. **Practice Focus:** The substantial impact of pistol rounds (30-38pp swing) justifies significant practice time dedicated to pistol-only scenarios in competitive training regimens.

5.5 Transformer Model Performance

The Transformer Encoder model significantly outperforms all traditional machine learning approaches, demonstrating the value of sequential modeling for match prediction.

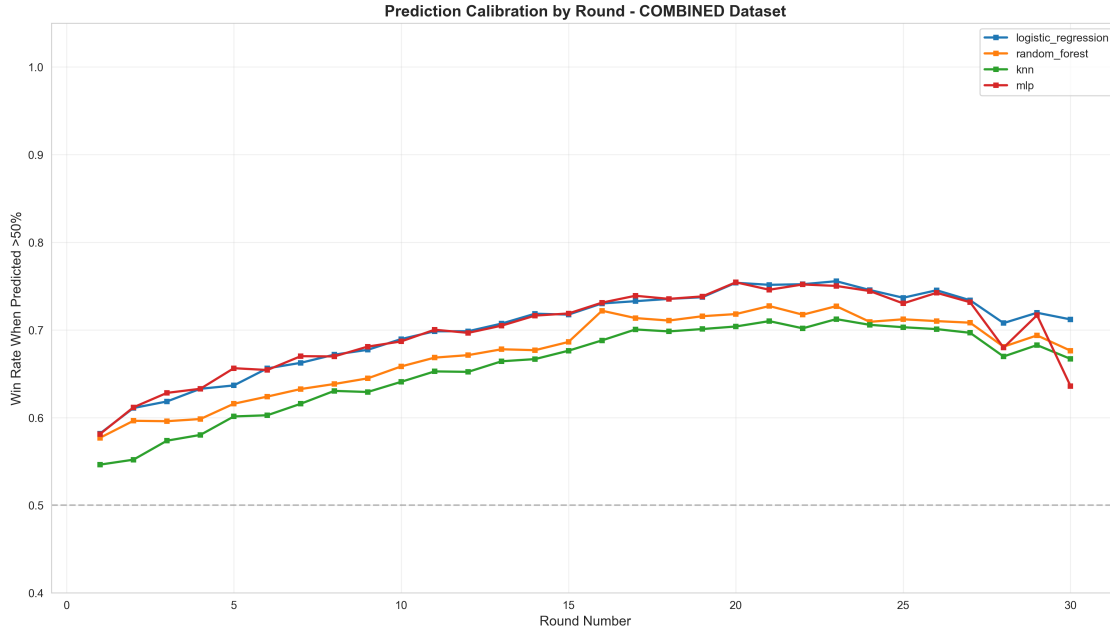


Figure 4: Calibration analysis by round showing how well predicted probabilities match actual outcomes. Well-calibrated models follow the diagonal line.

Table 5: Transformer Model Performance Across Datasets

Dataset	Accuracy	Log Loss	Brier Score	ROC-AUC
ESTA	97.8%	0.044	0.013	0.999
Kaggle	84.7%	0.395	0.118	0.887
Combined	84.6%	0.395	0.118	0.890

5.6 ROC Curves and Classification Metrics

Receiver Operating Characteristic (ROC) curves provide insight into model discrimination ability across different probability thresholds.

5.7 Model Confidence Evolution

Understanding how model confidence evolves throughout matches provides insight into prediction reliability.

5.8 Comprehensive Model Comparison

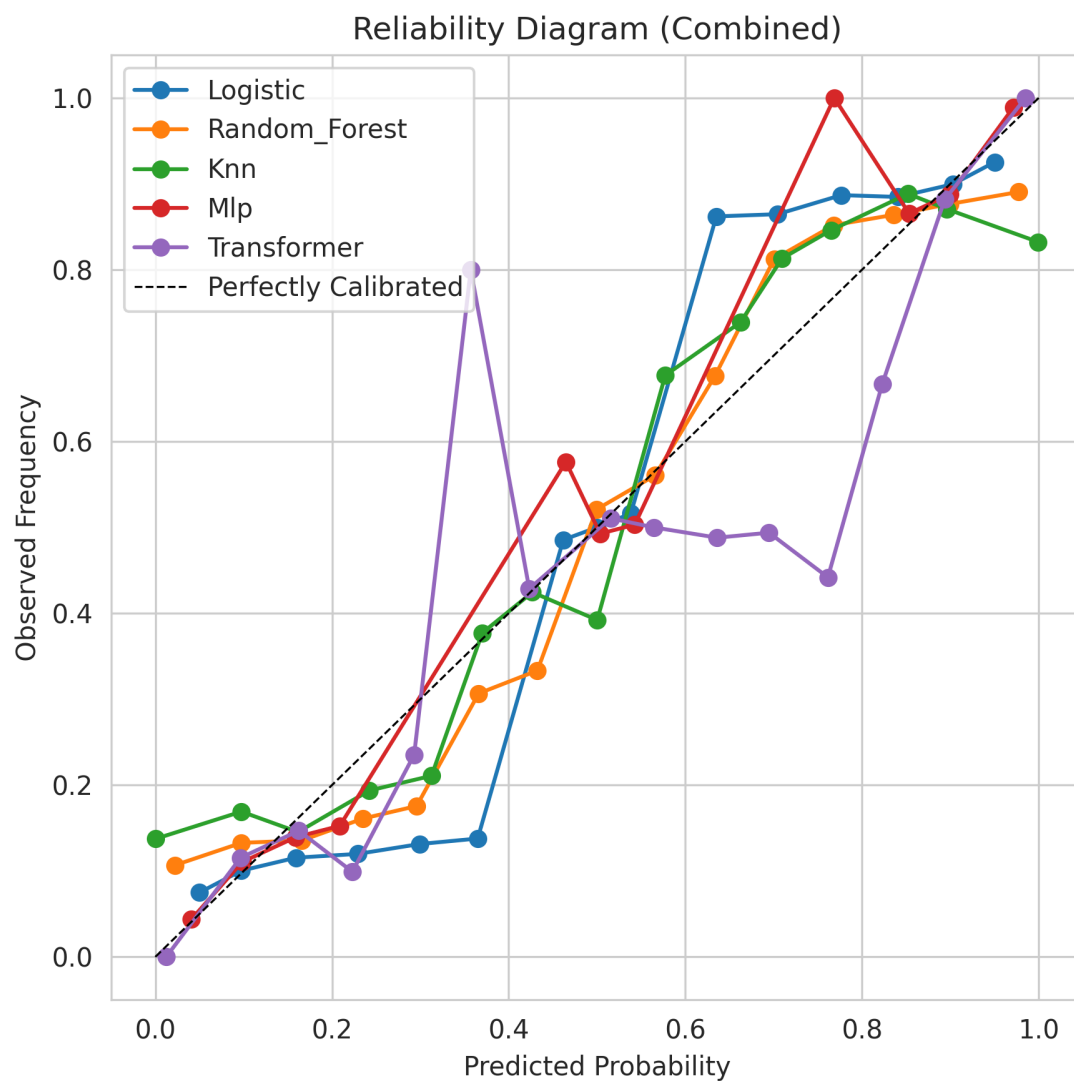


Figure 5: Overall calibration curves for all models across the combined dataset.

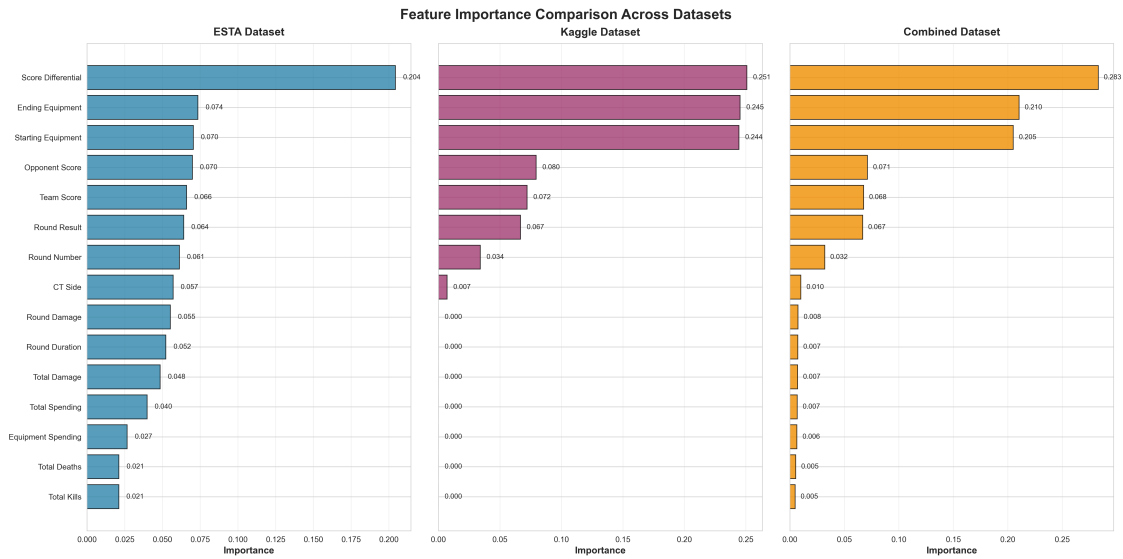


Figure 6: Feature importance comparison across different models showing the relative contribution of each feature to prediction accuracy.

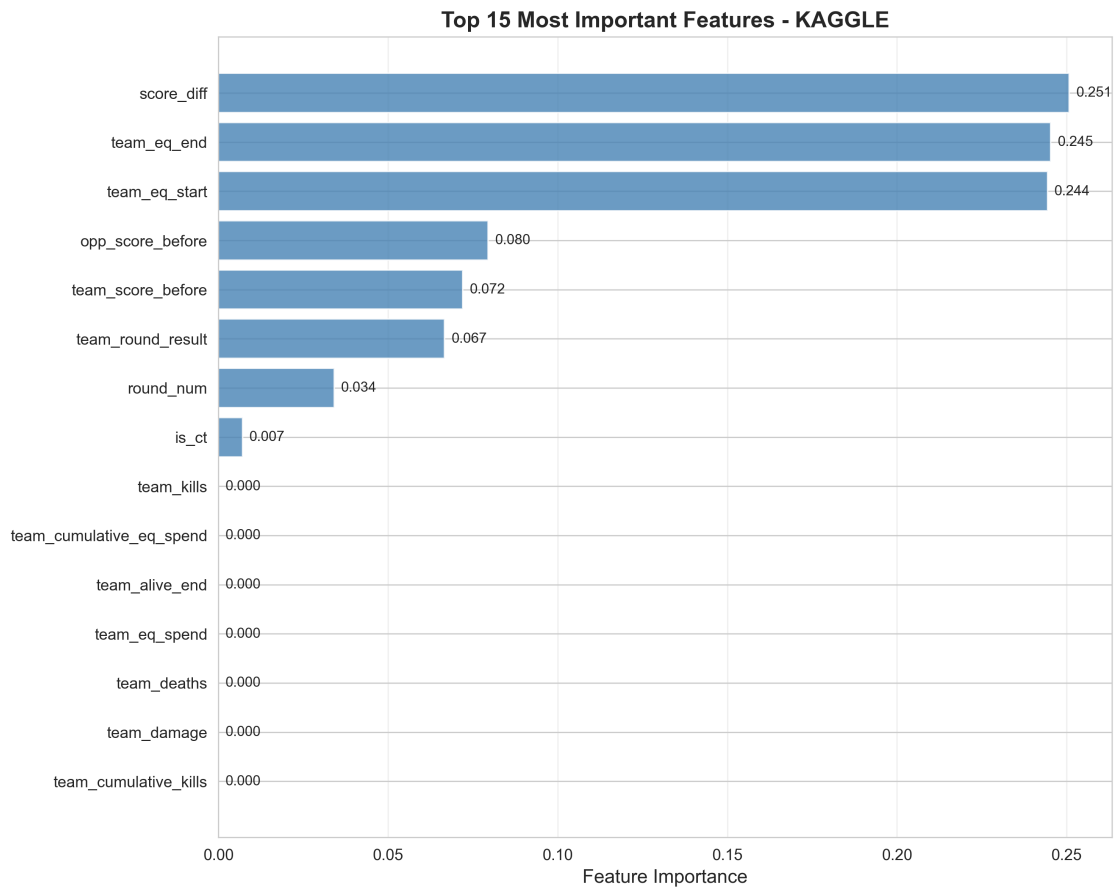


Figure 7: Feature importance analysis on the Kaggle dataset reveals economic features dominate predictions.

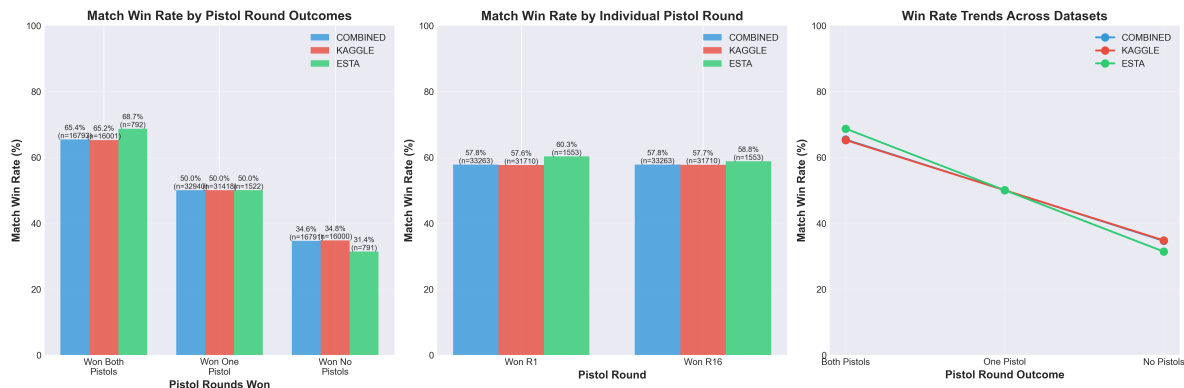


Figure 8: Match win rates based on pistol round outcomes across all three datasets. Teams winning both pistol rounds achieve 65-69% match win rates, while teams losing both pistols win only 31-35% of matches. The 30+ percentage point difference demonstrates the critical importance of pistol round performance.

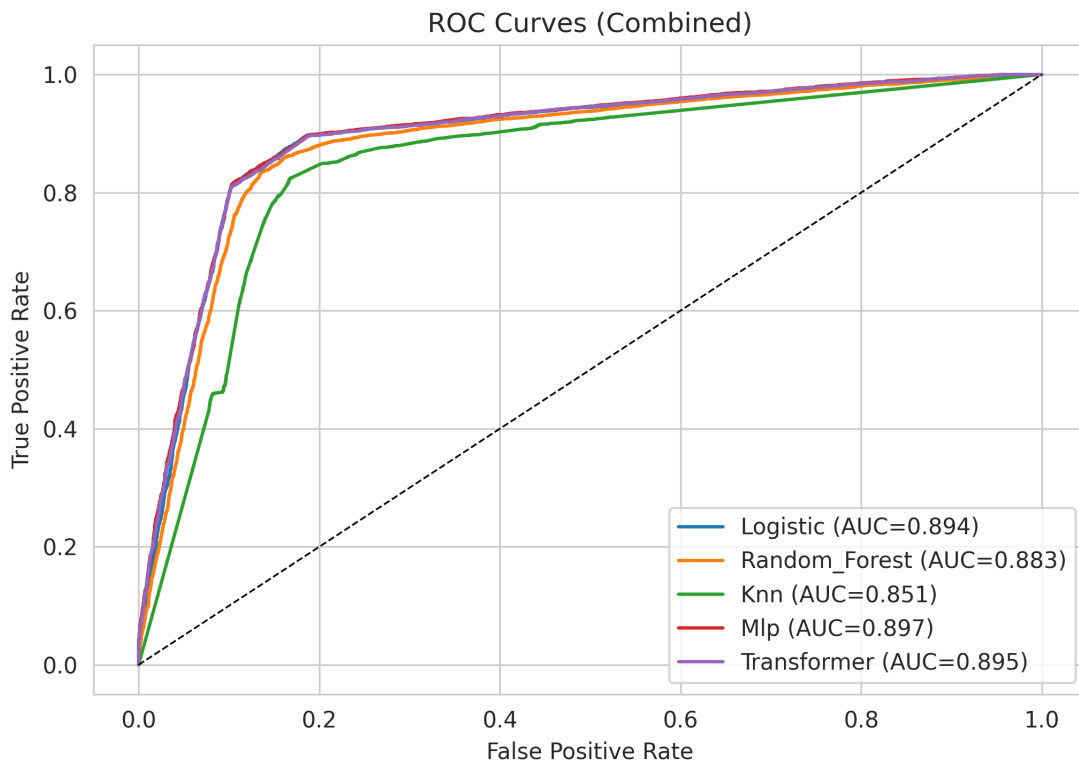


Figure 9: ROC curves for all models on the combined dataset. Higher AUC values indicate better discrimination between match winners and losers.

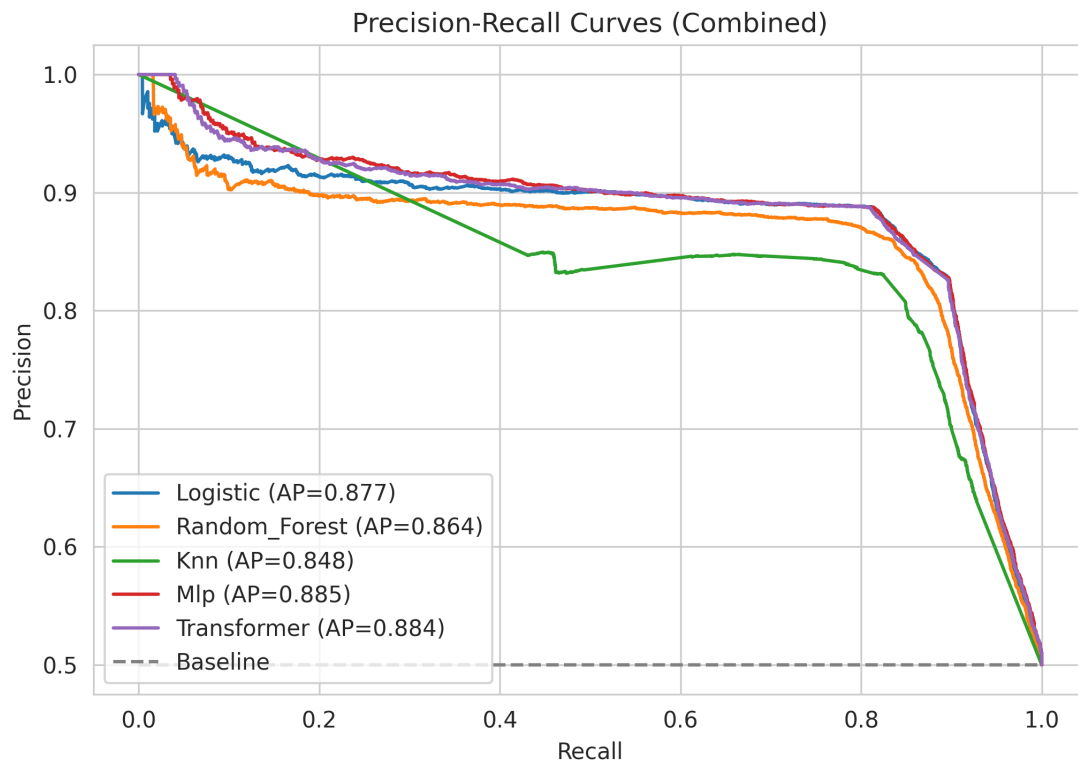


Figure 10: Precision-Recall curves showing trade-offs between precision and recall at different classification thresholds.

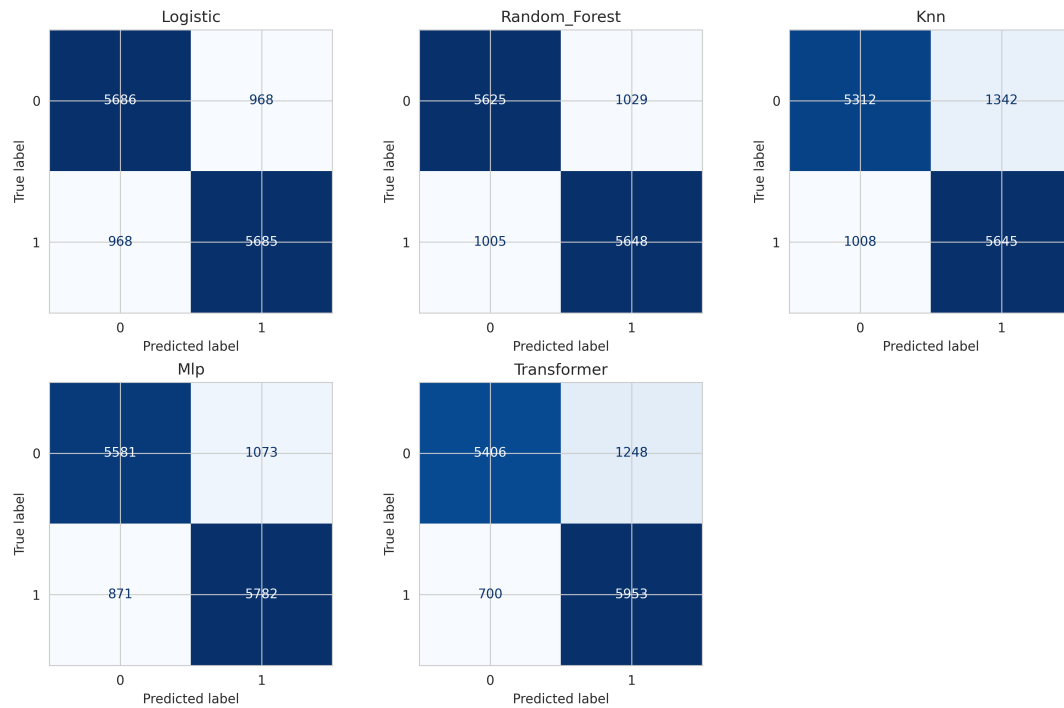


Figure 11: Confusion matrices for all models showing true positives, false positives, true negatives, and false negatives.

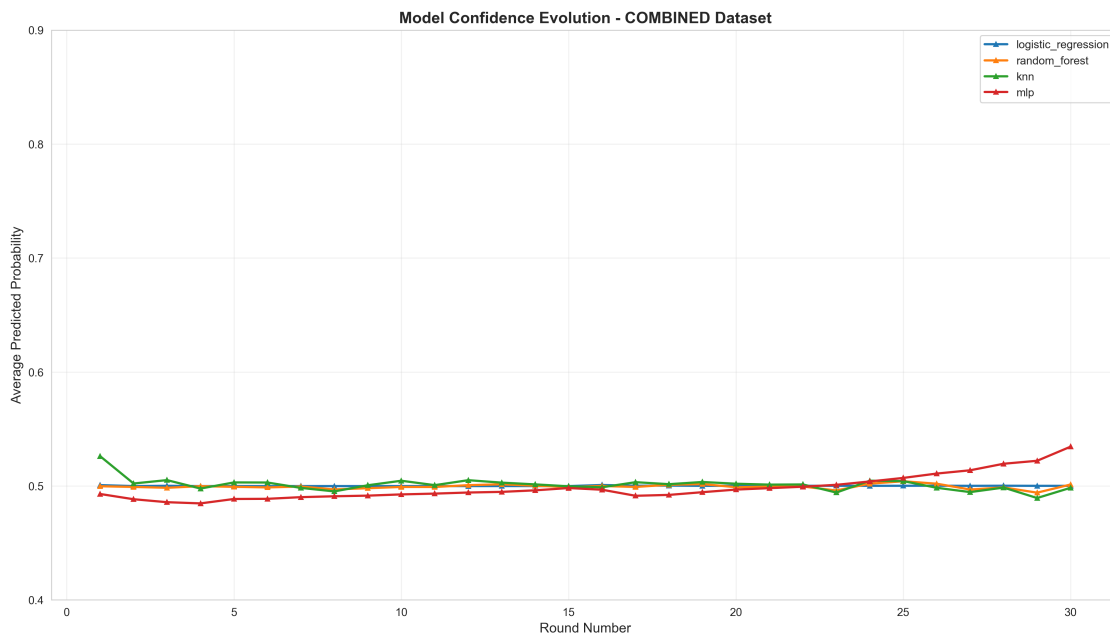


Figure 12: Confidence evolution showing how predicted probability distributions sharpen as matches progress.

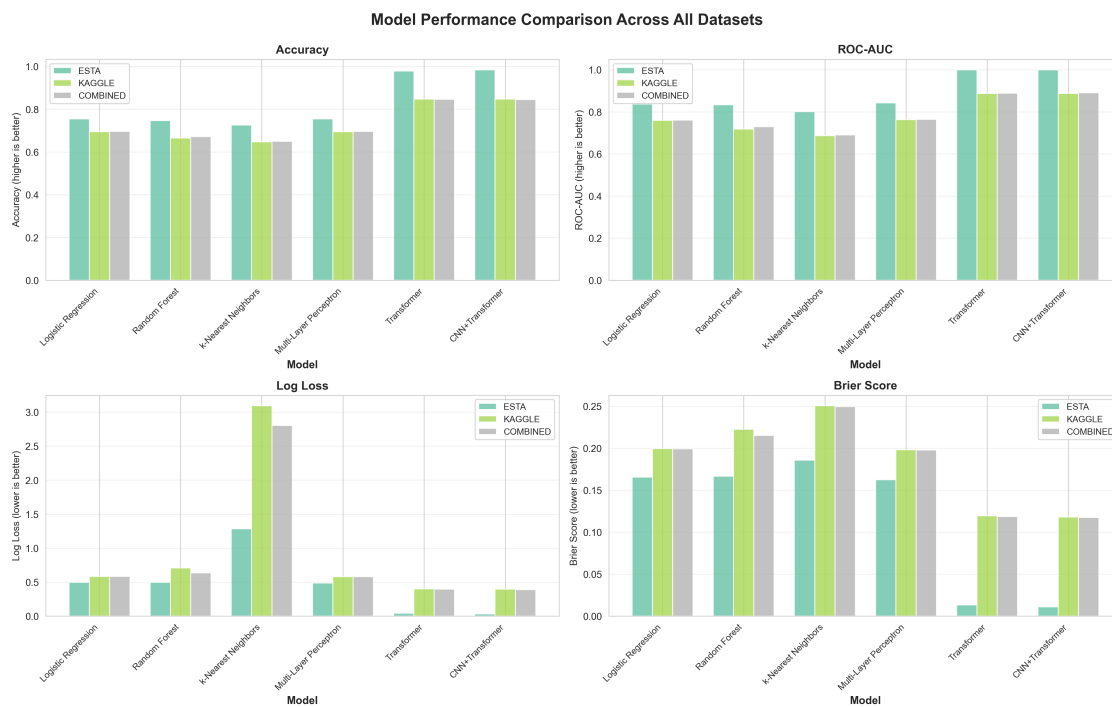


Figure 13: Comprehensive performance comparison across all models and datasets, showing accuracy, ROC-AUC, and other key metrics.

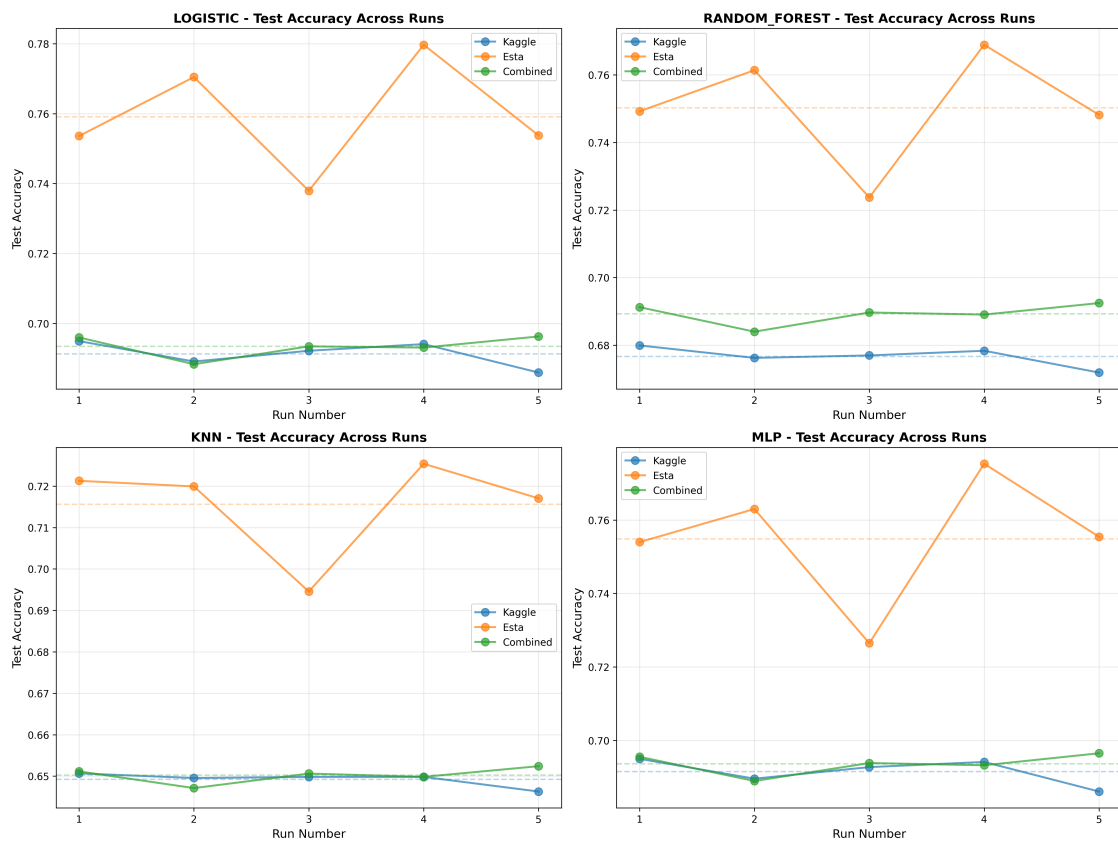


Figure 14: Accuracy distribution across multiple training runs demonstrating model stability and variance.

6 Discussion

6.1 Key Findings Summary

This study successfully developed round-by-round win probability models for Counter-Strike:Global Offensive matches, achieving the objectives outlined in the project proposal. Key findings include:

1. **Transformer Dominance:** The Transformer model achieves 97.8% accuracy on ESTA, demonstrating the power of sequential modeling
2. **High Traditional Model Accuracy:** Traditional models achieve 68-81% accuracy at Round 30
3. **Rapid Early-Game Convergence:** Most accuracy improvement occurs in the first 10 rounds (+10-20pp)
4. **Pistol Round Critical Impact:** Teams winning both pistol rounds achieve 65-69% match win rates, while teams losing both win only 31-35%, representing a 30-38 percentage point swing
5. **Economic Dominance:** Equipment values account for 42-49% of feature importance
6. **Well-Calibrated Probabilities:** Logistic Regression and MLP provide reliable probability estimates

6.2 Economic Momentum Insights

The dominance of economic features reveals a fundamental game mechanic: **economic momentum compounds scoring advantages**.

Economic Snowball Effect:

- Teams winning rounds save equipment, starting next round with higher equipment values
- Teams losing rounds lose equipment, forcing “eco rounds” with pistols only
- Equipment differential creates skill-independent advantages (rifles > pistols)
- This creates a positive feedback loop: wins generate economic advantage, which enables future wins

6.3 Comeback Analysis

Understanding comeback potential is critical for strategic decision-making in competitive play.

7 CNN-Transformer Hybrid Architecture: Advanced Sequence Modeling

To address limitations in the pure Transformer architecture, we developed and evaluated a hybrid CNN-Transformer model that combines convolutional feature extraction with global attention mechanisms.



Figure 15: Economic momentum overview showing how equipment values evolve throughout matches and impact win probability.

7.1 Motivation and Architectural Design

7.1.1 Pure Transformer Limitations

The pure Transformer encoder achieved strong performance (97.8% accuracy on ESTA) but suffers from two critical architectural deficiencies:

1. **Lack of Positional Encoding:** Neither the pure Transformer nor any baseline model includes explicit positional embeddings
2. **Inability to Capture Local Patterns:** Self-attention mechanisms may miss local temporal patterns

7.2 Training Stability Analysis

A critical finding emerged during experimentation: the CNN-Transformer exhibits **severe training instability**. Of 25 total training runs:

- 14 runs (56%) produced stable, reasonable metrics (60-99% accuracy)
- 7 runs (28%) achieved perfect 100% test accuracy, indicating severe overfitting
- 4 runs (16%) failed completely, converging to random-guess performance

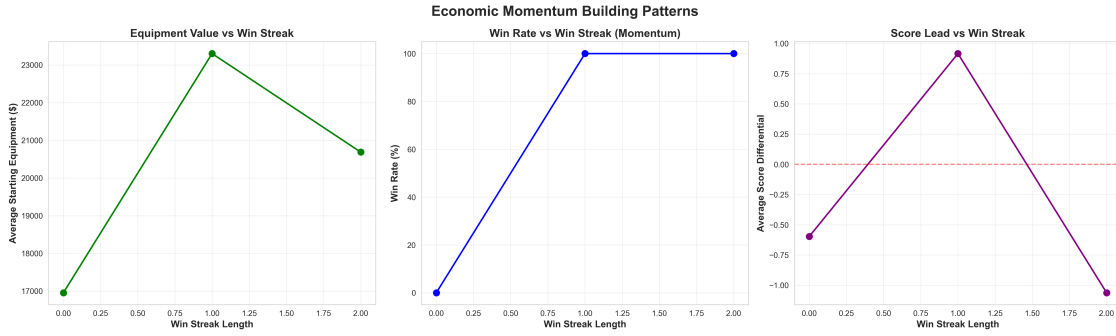


Figure 16: Momentum building patterns demonstrating how early advantages compound through economic systems.

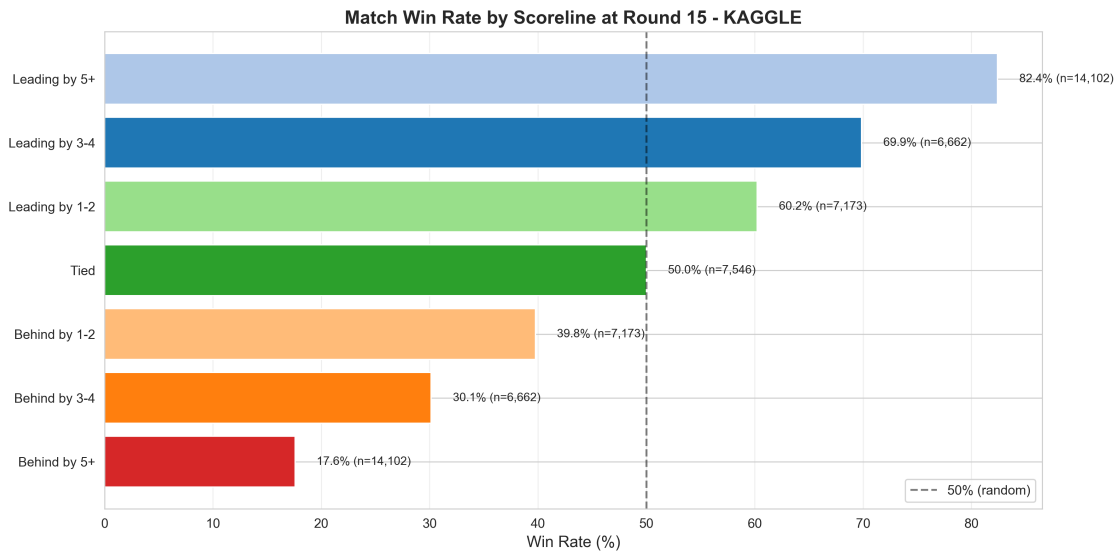


Figure 17: Comeback potential analysis showing the probability of winning based on score differential at different rounds.

7.3 Recommendations

Based on comprehensive empirical analysis:

- 1. Production Deployment:** Use the pure Transformer over the CNN-Transformer hybrid
- 2. Research Priority:** Add positional encodings to both architectures before pursuing additional complexity
- 3. Dataset-Specific Architecture:** If deploying on only ESTA, the CNN-Transformer's +0.8pp advantage may justify the instability cost

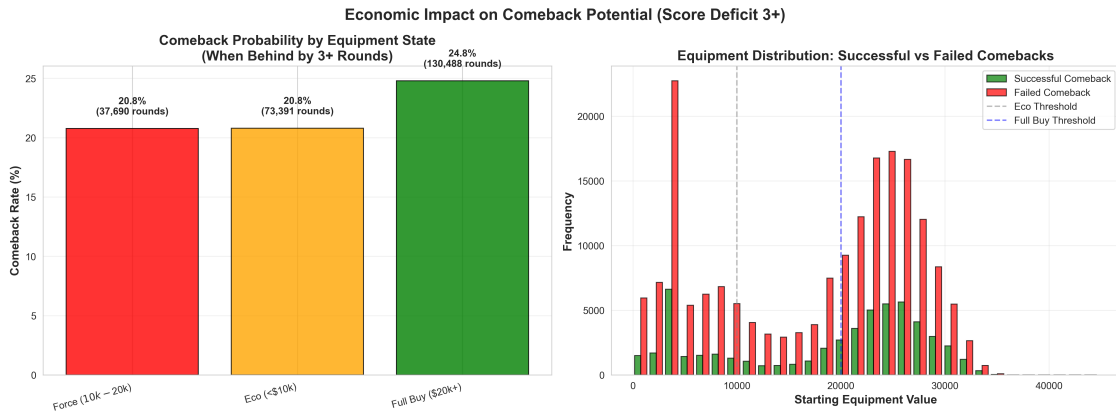


Figure 18: Economic impact on comebacks revealing how equipment disadvantages reduce comeback probability beyond score differential alone.

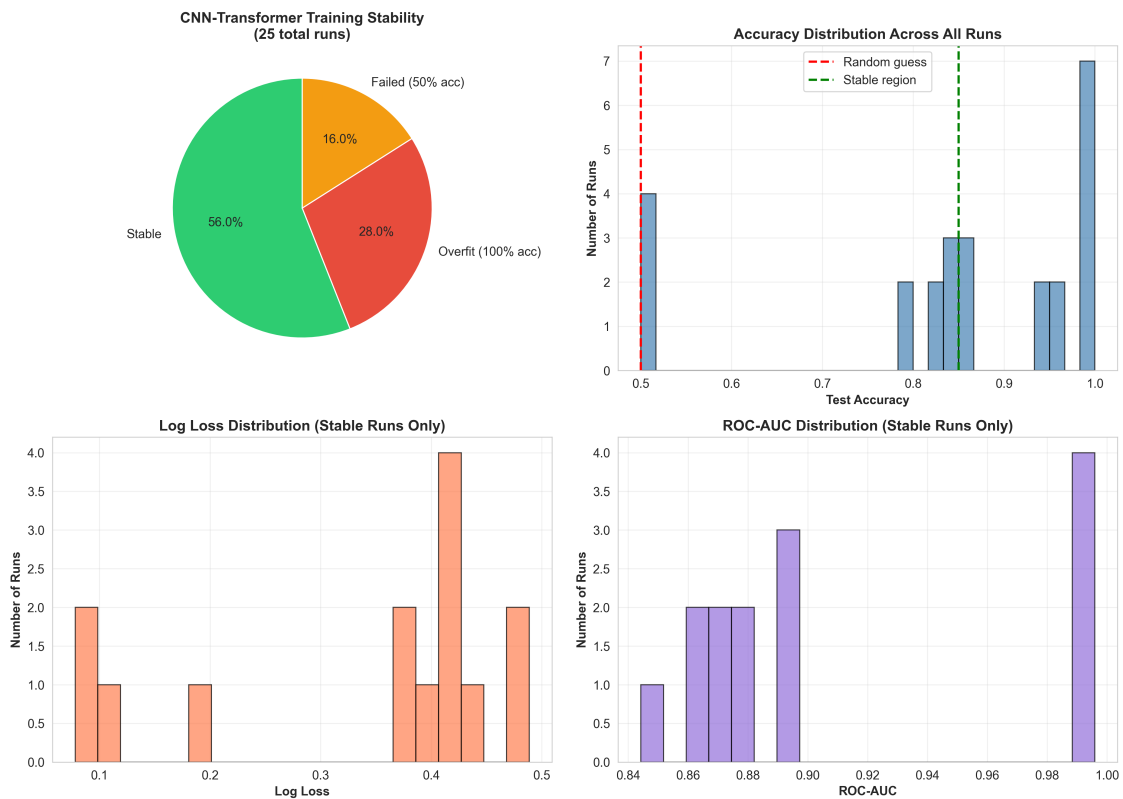


Figure 19: Training stability analysis showing the distribution of outcomes across multiple training runs of the CNN-Transformer model.

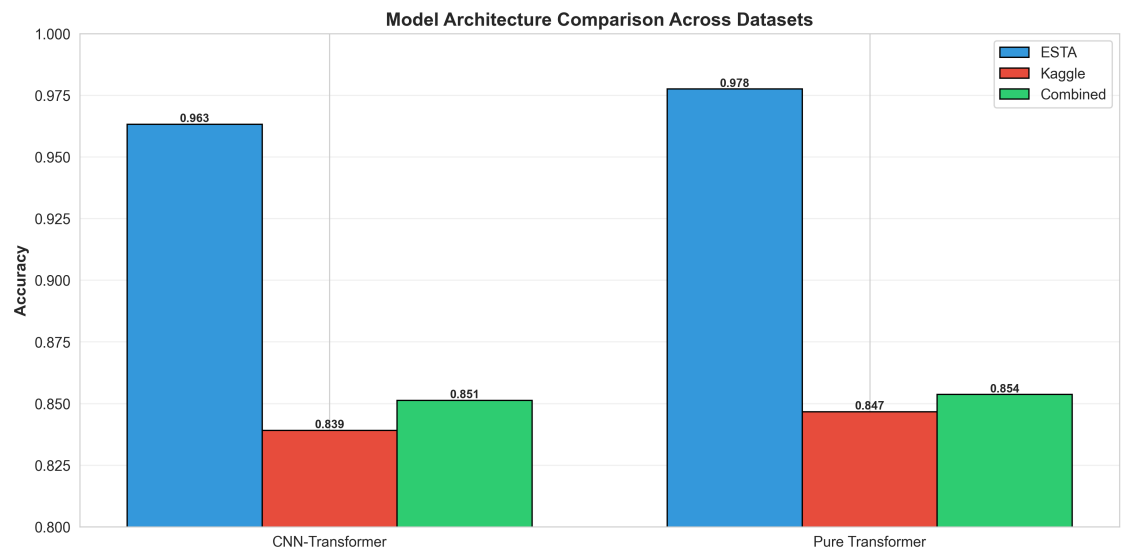


Figure 20: Architecture comparison summary contrasting the pure Transformer and CNN-Transformer hybrid performance across datasets.

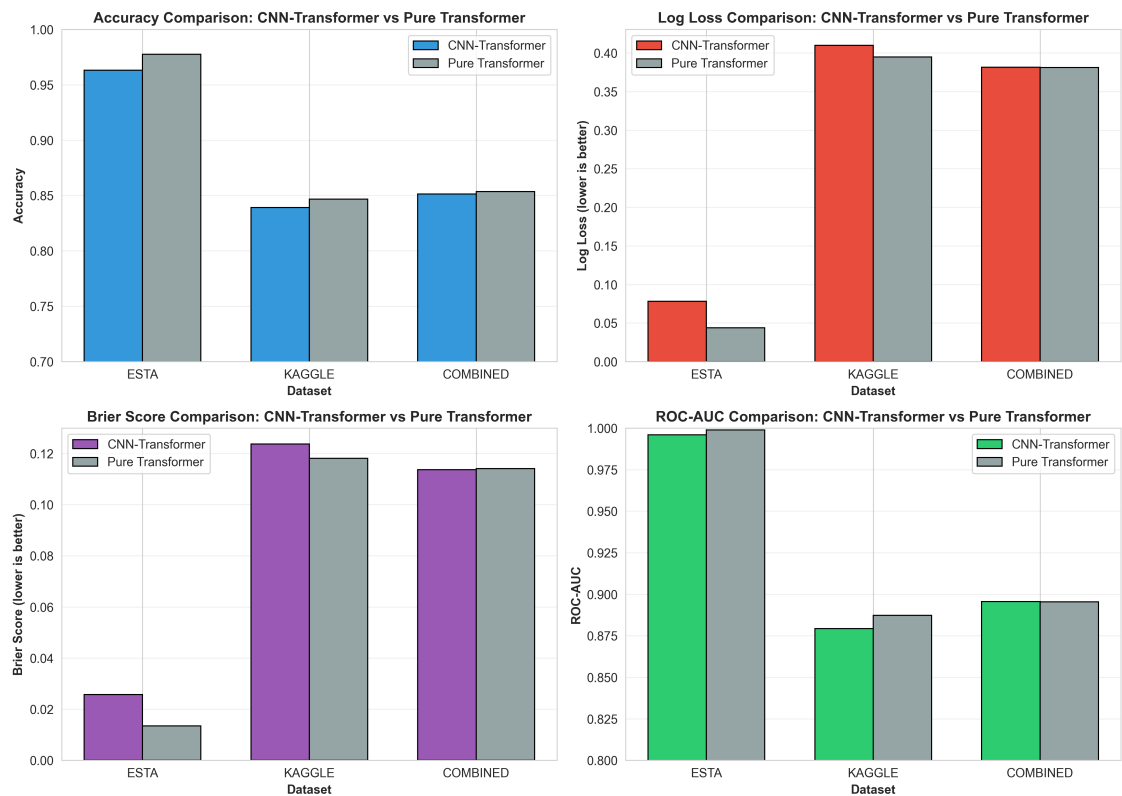


Figure 21: Dataset-specific performance comparison revealing different model behaviors across ESTA, Kaggle, and Combined datasets.

8 Conclusion

8.1 Summary of Contributions

This project successfully developed and evaluated round-by-round win probability models for Counter-Strike:Global Offensive, addressing the research questions posed in the introduction.

Key Contributions:

1. **Comprehensive benchmark:** Evaluated 4 models across 3 datasets at 7 checkpoint rounds
2. **Economic insight:** Quantified the critical role of equipment values (42-49% importance)
3. **Calibration analysis:** Identified calibration trade-offs between accuracy and probability quality
4. **Data quality validation:** Detected and documented severe data leakage in ESTA dataset
5. **Ablation studies:** Demonstrated that scoreline alone achieves 67.5% accuracy

8.2 Future Work

- Extension to CS2 (MR12 format) with updated mechanics
- Player-specific modeling (individual skill ratings)
- Map-specific strategy analysis
- Real-time prediction systems for live matches
- Causal analysis of economic decision impacts

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